

Batteries for IoT Systems

Why Batteries Matter in IoT

Most IoT devices are **not plugged into a wall** — they run on batteries.

Real-world IoT scenarios:

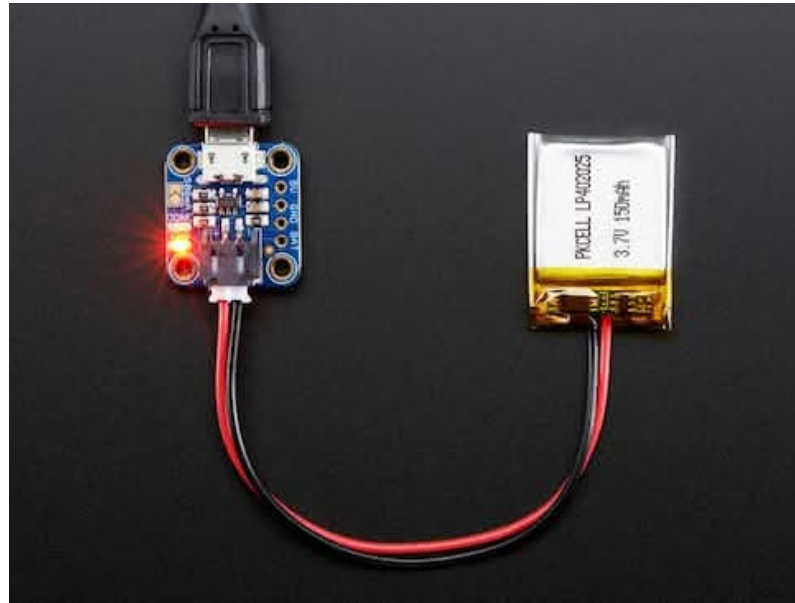
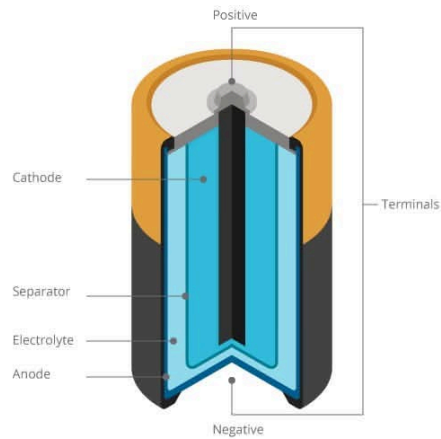
- Soil moisture sensor in a field → **AA batteries**
- Smart door lock → **Li-ion rechargeable pack**
- Wearable health monitor → **LiPo pouch cell**
- Emergency GPS tracker → **coin cell (CR2032)**

Questions every IoT engineer must answer:

1. What voltage does my circuit need?
2. How long must the battery last?
3. How is the battery recharged?

Battery Voltage — The Basics

A battery produces voltage through a **chemical reaction** between two electrodes (cathode and anode) in an electrolyte.

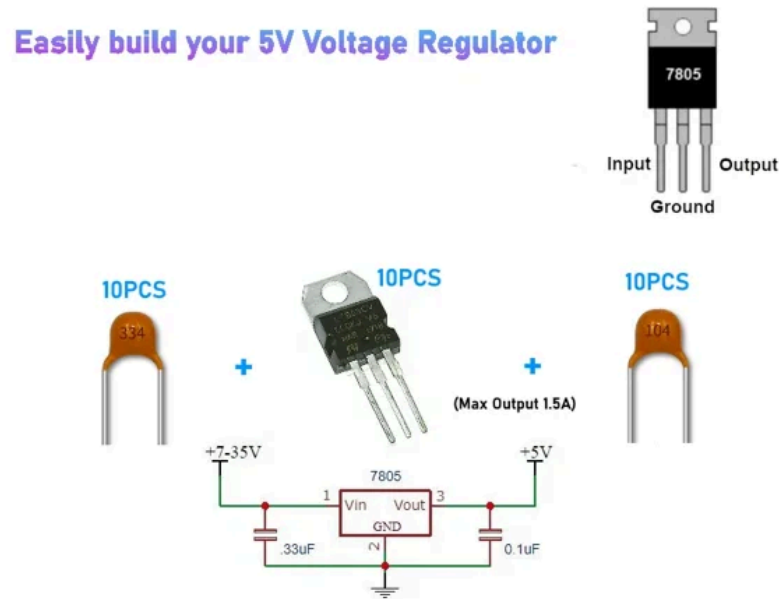


- **Cell voltage** = voltage of a single electrochemical cell
- **Battery voltage** = cells in series → voltages add up

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1 cell:    3.7V (Li-ion)
2 cells:   7.4V (2S Li-ion pack)
3 cells:  11.1V (3S Li-ion pack)
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Always check: does your MCU/sensor run at 3.3V or 5V? You may need a regulator.

- Two capacitors and 7805 regulator makes a stable 5V supply from a 7.4V (or above 5V) battery pack.



Common Battery Voltages by Chemistry

Chemistry	Nominal Voltage	Min (discharged)	Max (full)
Alkaline (AA/AAA)	1.5V	0.8V	1.65V
NiMH	1.2V	1.0V	1.45V
Li-ion (18650)	3.6–3.7V	3.0V	4.2V
LiPo	3.7V	3.0V	4.2V
LiFePO ₄ (LFP)	3.2V	2.5V	3.65V
CR2032 (coin)	3.0V	2.0V	3.0V

Important for IoT: MCUs like ESP32 (3.3V) need regulation when using 4.2V Li-ion at full charge.

Battery Technologies

Alkaline (AA, AAA, C, D)

- Cheap, widely available
- **Not rechargeable** (standard type)
- Good for low-power sensors (years of life)
- Voltage drops gradually → easy to estimate remaining charge



Coin Cell (CR2032, CR2025, etc.)

- Small, lightweight, 3V lithium primary battery
- **Not rechargeable** (standard CR type)
- Very low current capability (μA – mA range)
- Ideal for RTC, BLE beacons, small IoT nodes
- Flat discharge curve $\rightarrow$ harder to estimate remaining life



NiMH (Nickel-Metal Hydride)



- Rechargeable, same size as alkaline
- **1.2V per cell** (slightly less than alkaline 1.5V)
- Good cycle life (~500–1000 charges)
- Used in older IoT and hobby devices

Practical IoT use:

Alkaline AA $\times 2 = 3V$ (good for 3.3V MCU via boost converter)

NiMH AA $\times 3 = 3.6V$ (workable for 3.3V MCU)

Lithium-Ion (Li-ion)

Li-ion is the dominant rechargeable chemistry for modern IoT.

Property	Value
Nominal voltage	3.6–3.7V
Full charge	4.2V
Minimum voltage	3.0V (never go below!)
Energy density	~250 Wh/kg
Cycle life	300–500 cycles
Temperature range	0°C to 45°C (charging)

Common Form Factors

- **18650** — cylindrical, 18mm × 65mm, used in laptop/power bank
- **21700** — larger, higher capacity (used in EVs, power tools)
- **Pouch** — flat, flexible packaging



21700 BATTERY
PROTECTED
BUTTON TOP



Battery Technologies — LiPo

LiPo (Lithium Polymer) — same chemistry as Li-ion, different packaging.

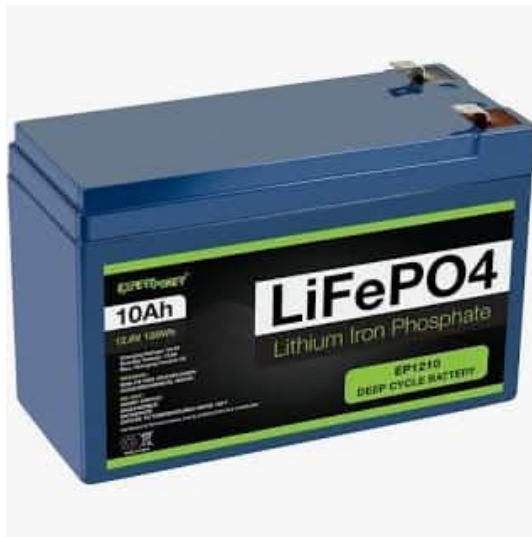


Feature	Li-ion	LiPo
Shape	Fixed cylinder	Any shape
Weight	Heavier	Lighter
Safety	More stable	Puffy if damaged
IoT use	Power banks, high capacity	Wearables, drones

Warning: LiPo cells can swell or catch fire if overcharged or punctured. Always use a protection circuit.

LiFePO_4 (LFP)

LiFePO_4 (Lithium Iron Phosphate) — the safest lithium chemistry.



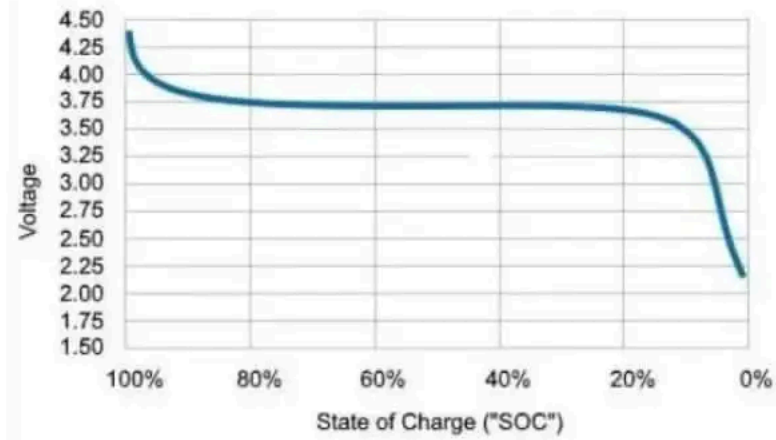
Why IoT engineers love LFP:

- **3.2V nominal** → nearly flat discharge curve
- Very safe — no thermal runaway
- **2000–4000 charge cycles** (far more than Li-ion)
- Operates well in wide temperature range

Great for solar IoT systems that charge/discharge daily.

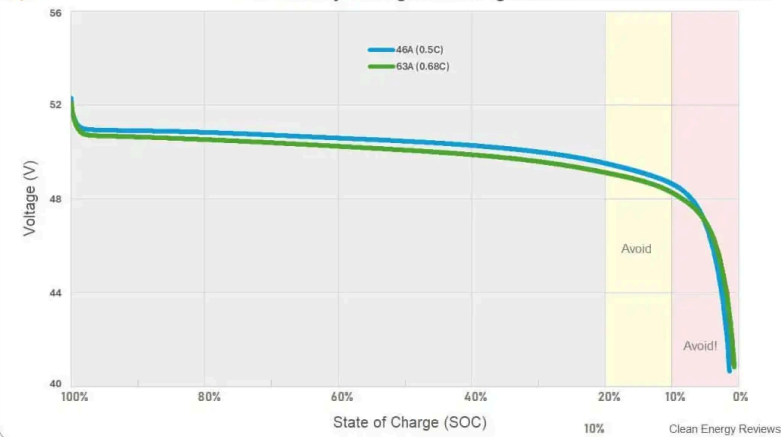


Typical Li-ion Discharge Voltage Curve



LFP Battery Voltage Discharge Curve*

* PPE Eco4847P
LFP Battery



The Theory of Charging a Battery

Charging is the **reverse of discharging** — we force current into the battery to restore chemical energy.

Key principle: you cannot just apply a fixed voltage!

Too much voltage → **overcharge** → heat, gas, fire

Too little voltage → **undercharge** → short battery life

Solution: Charge Controller IC manages the charging process automatically.

Li-ion / LiPo Charging: CC-CV Method

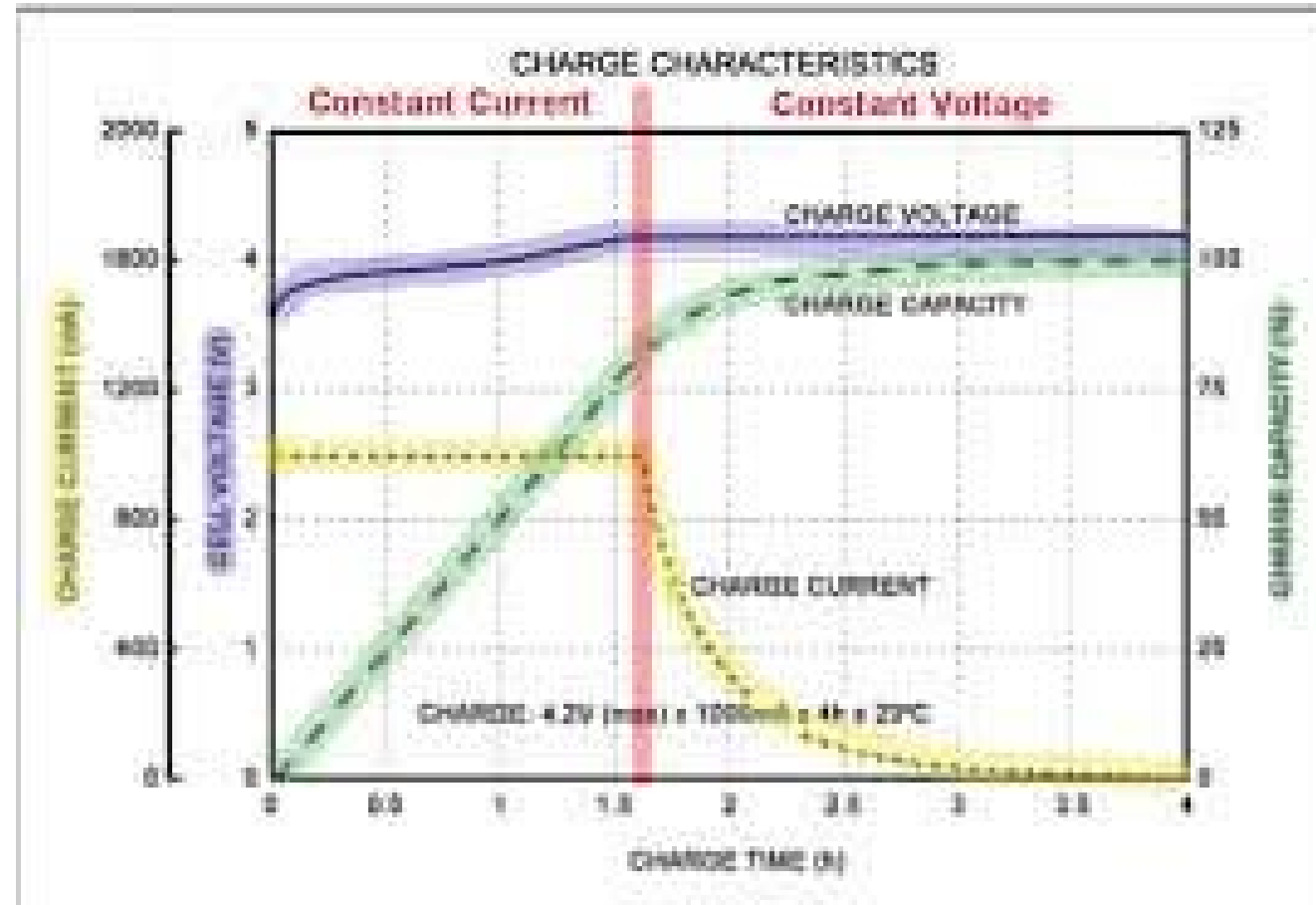
The standard charging method for lithium batteries:

Stage 1: Constant Current (CC)

- └ Apply fixed current (e.g., 1A)
- └ Voltage rises gradually
- └ Until voltage reaches 4.2V

Stage 2: Constant Voltage (CV)

- └ Hold voltage at 4.2V
- └ Current gradually decreases
- └ Stop when current $< 0.05C$ (e.g., 50mA for 1000mAh battery)



C-rate — Charging Current Relative to Capacity

C-rate = charge current / battery capacity

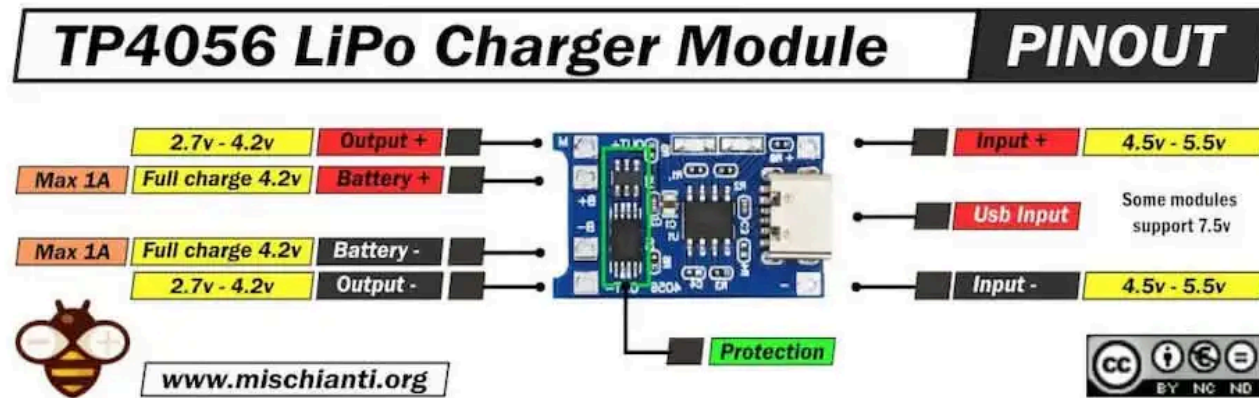
- Example: 1C for a 1000mAh battery = 1000mA (1A)
- 0.5C = 500mA, 0.1C = 100mA
- Higher C-rate → faster charge but more stress on battery
- For IoT, 0.5C or lower is often recommended for longevity
- Some batteries specify max C-rate (e.g., 2C) — never exceed it!

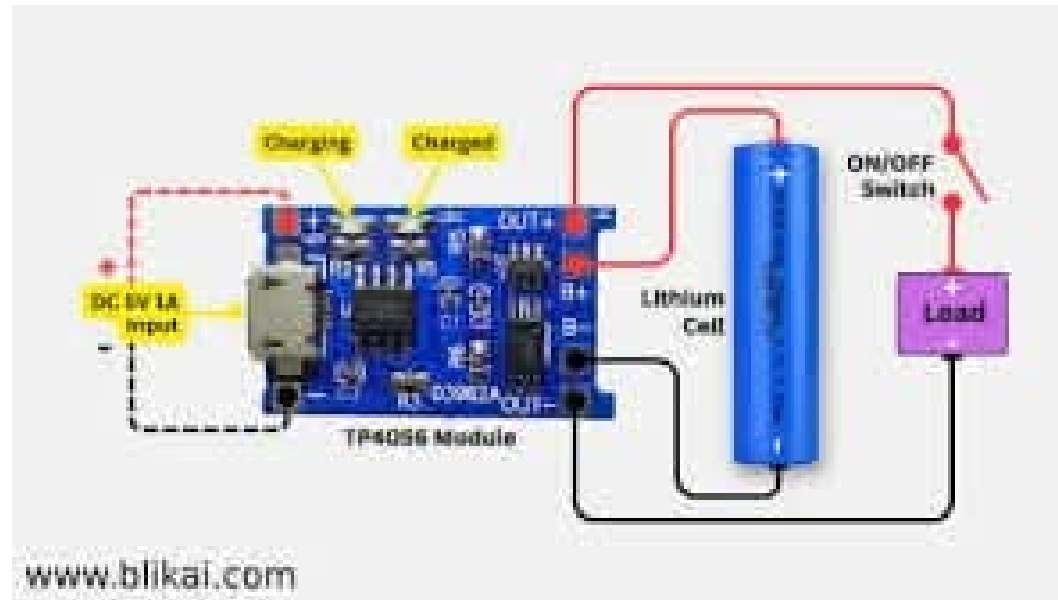
Popular Li-ion Charge Controller ICs

Chip	Max Current	Input	Features	Cost
TP4056	1A	5V USB	CC-CV, LED indicators	~\$0.20
MCP73831	500mA	3.75–6V	Tiny SOT-23 package	~\$0.50
BQ24079	1.5A	USB/DC	System load + charge	~\$1.50
MAX1555	280mA	USB or DC	Dual input auto-select	~\$2.00
IP5306	2A	USB	Charge + boost + LED	~\$0.80

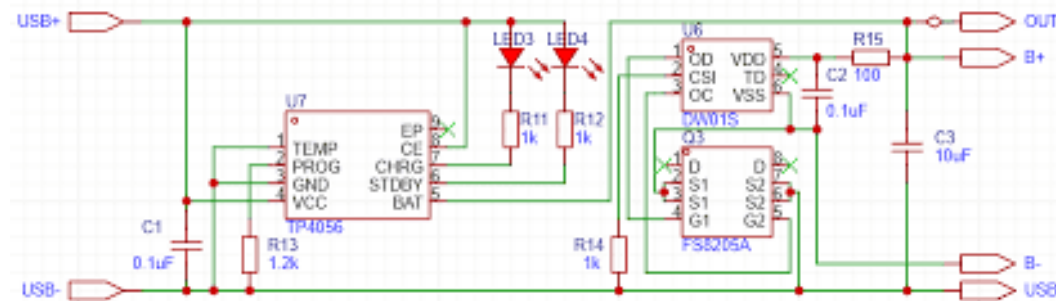
TP4056 — The Ubiquitous Li-ion Charger for IoT

USB 5V → TP4056 → Li-ion battery
└→ Load output (with protection module)





- This board has protection circuitry (overcharge, over-discharge) with DW01 and dual MOSFET.
- This board charges the battery and provides direct (unregulated) battery output to the load.
- When not charging, the load is powered directly from the battery through the protection circuit.



R13 (R_prog) = programming resistor to set charge current (I_charge)

$$I_{\text{charge}} = 1200 / R_{\text{prog}} \text{ (in ohms)}$$

$$R_{\text{prog}} = 1.2\text{k}\Omega \rightarrow I = 1000\text{mA} \text{ (1A)}$$

$$R_{\text{prog}} = 2.0\text{k}\Omega \rightarrow I = 600\text{mA}$$

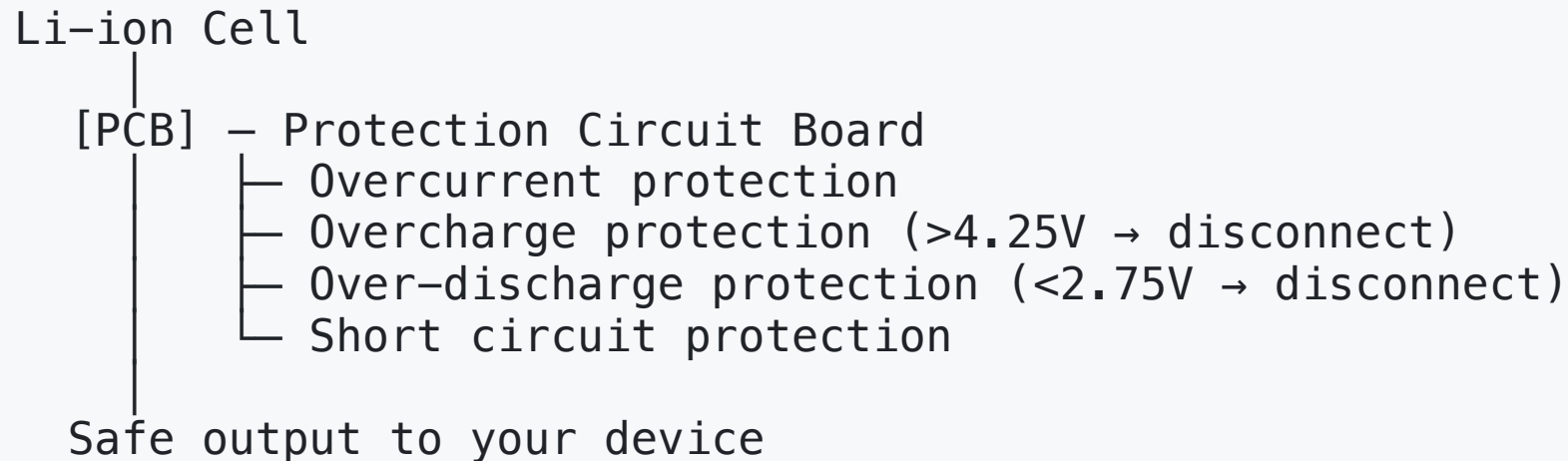
$$R_{\text{prog}} = 3.0\text{k}\Omega \rightarrow I = 400\text{mA}$$

For small LiPo cells (<500mAh), use lower current to protect the battery.

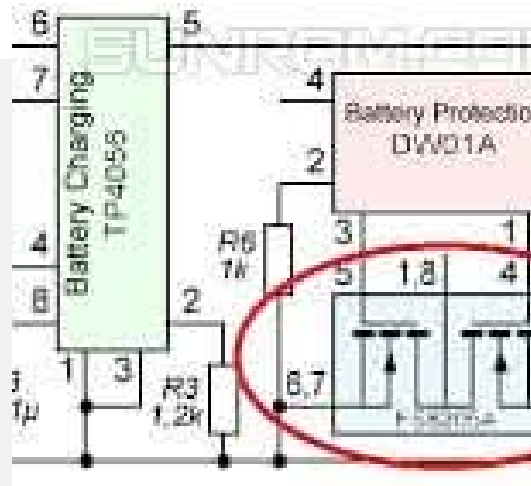
Battery Protection Circuit (PCB/PCM)

Bare Li-ion cells **have no protection** — you must add a protection circuit.

- PCB = Protection Circuit Board
- PCM = Protection Circuit Module



- Li-ion batteries sold for consumer use usually include PCB
- **Bare cells** (used in DIY packs) need external protection
- Many charge ICs (including TP4056) include protection logic (e.g., DW01A + FS8205)



Mimicking Notebook Charging Behavior

When you plug in a laptop, it charges the battery while powering the system.

- The system is powered directly from the adapter through a power-path management circuit.
- The battery is charged separately.
- The output to the system is regulated and stable.

This is NOT how TP4056 works.

- TP4056 only charges the battery.
- The load is powered directly from the battery (through the protection circuit).
- The output is not regulated — it follows the battery voltage.
- There is no true power-path management.

Power-Path Management for IoT

- TI BQ24074
- Microchip MCP73871



NiMH Charging — Different Rules

NiMH batteries use a **different charging algorithm** from Li-ion.

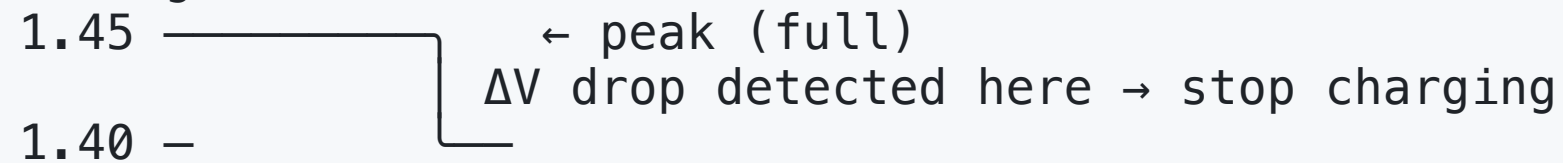
Method 1: Slow Charge (C/10)

- Charge at 1/10th of capacity for 14–16 hours
- Simple, no controller needed (just a resistor + timer)
- Safe for most NiMH batteries

Method 2: Fast Charge (1C) with ΔV Detection

NiMH voltage during fast charge:

Voltage



Charge controller watches for voltage PEAK then small DROP ($\Delta V \approx -5\text{mV/cell}$)

Never use Li-ion charger on NiMH — different voltage profiles!